



AQ9.2: Activity Questions 2 - Not Graded



This assignment will not be graded and is only for practice.

Partial Derivatives:

Level 1:

What is the rate of change of $f(x, y, z) = 5x^2 - 6xy + 3z^2 - 2yz$ at $(1, 2, -1)$ with respect to z ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -10

1 point

1 point

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined as $f(x, y) = x^3 + 3xy$. Identify the correct option with respect to the partial derivatives of f .

☐

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

☐

$$\frac{\partial f}{\partial x}(0, 0) = \frac{\partial f}{\partial y}(0, 0) \quad \frac{\partial^2 f}{\partial x \partial y}(0, 0) = \frac{\partial^2 f}{\partial y \partial x}(0, 0)$$

☐

$$\frac{\partial f}{\partial x}(1, 1) = \frac{\partial f}{\partial y}(1, 1) \quad \frac{\partial^2 f}{\partial x \partial y}(1, 1) = \frac{\partial^2 f}{\partial y \partial x}(1, 1)$$

☐

$$\frac{\partial f}{\partial x}(1, 0) = \frac{\partial f}{\partial y}(1, 0) \quad \frac{\partial^2 f}{\partial x \partial y}(1, 0) = \frac{\partial^2 f}{\partial y \partial x}(1, 0)$$

☐

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} + 3 \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} + 3$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\frac{\partial f}{\partial x}(0, 0) = \frac{\partial f}{\partial y}(0, 0) \quad \frac{\partial^2 f}{\partial x \partial y}(0, 0) = \frac{\partial^2 f}{\partial y \partial x}(0, 0)$$

$$\frac{\partial f}{\partial x}(1, 0) = \frac{\partial f}{\partial y}(1, 0) \quad \frac{\partial^2 f}{\partial x \partial y}(1, 0) = \frac{\partial^2 f}{\partial y \partial x}(1, 0)$$

1 point

Let $f(x, y) = \frac{xy}{x^2 + y}$. Identify the correct options with respect to the partial derivatives of f .

☐

$$\frac{\partial f}{\partial x} \text{ and } \frac{\partial f}{\partial y} \text{ are not defined at } (0, 0)$$

☐

$$\frac{\partial f}{\partial x}(1, 1) = 0, \frac{\partial f}{\partial y}(1, 1) = 1/4 \quad \frac{\partial^2 f}{\partial x \partial y}(1, 1) = 0, \frac{\partial^2 f}{\partial y \partial x}(1, 1) = 1/4$$

☐

$$\frac{\partial f}{\partial x}(1, 1) = 0, \frac{\partial f}{\partial y}(1, 1) = 0 \quad \frac{\partial^2 f}{\partial x \partial y}(1, 1) = 0, \frac{\partial^2 f}{\partial y \partial x}(1, 1) = 0$$

☐

$$\frac{\partial f}{\partial x}(1, 0) = 0, \frac{\partial f}{\partial y}(1, 0) = 1 \quad \frac{\partial^2 f}{\partial x \partial y}(1, 0) = 0, \frac{\partial^2 f}{\partial y \partial x}(1, 0) = 1$$

No, the answer is incorrect.

Score: 0

Accepted Answers:



$\frac{\partial f}{\partial x} \partial x \partial f$ and $\frac{\partial f}{\partial y} \partial y \partial f$ are not defined at $(0,0)$

$$\frac{\partial f}{\partial x}(1,1) = 0, \frac{\partial f}{\partial y}(1,1) = 1/4, \partial x \partial f(1,1) = 0, \partial y \partial f(1,1) = 1/4$$

$$\frac{\partial f}{\partial x}(1,0) = 0, \frac{\partial f}{\partial y}(1,0) = 1, \partial x \partial f(1,0) = 0, \partial y \partial f(1,0) = 1$$

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined as $f(x, y) = x^2 + 4xy - y^2$. Suppose $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} \partial x \partial f = \partial y \partial f$ at $(12, k)$. Find the value of k

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 4

1 point

Level 2:

1 point

Suppose $f(x, y, z) = xy + yz + zx - 3xyz$ is a multivariable function defined on domain $D \subset \mathbb{R}^3$. Which of the following is(are) correct?

☐

f is a scalar valued multivariable function.

☐

The rate of change of the function f at $(0,1,1)$ with respect to x is 1.

☐

The rate of change of the function f at $(2,2,2)$ with respect to x is same as the rate of change of the function f at $(2,2,2)$ with respect to z .

☐

The rate of change of the function f at $(1,3,5)$ with respect to y is -9.

No, the answer is incorrect.

Score: 0

Accepted Answers:

f is a scalar valued multivariable function.

The rate of change of the function f at $(2,2,2)$ with respect to x is same as the rate of change of the function f at $(2,2,2)$ with respect to z .

The rate of change of the function f at $(1,3,5)$ with respect to y is -9.

1 point

Suppose $f: D \rightarrow \mathbb{R}$ is a multivariable function defined on domain $D \subset \mathbb{R}^2$ and also satisfying the following conditions:

$$(i) \frac{\partial f}{\partial x} = 6x + 4xy + 3y, \partial x \partial f = 6x + 4xy + 3y$$

$$(ii) \frac{\partial f}{\partial y} = 2x^2 + 3x - 3y, \partial y \partial f = 2x^2 + 3x - 3y$$

Which of the following functions can not be f ?

☐

$$f(x, y) = 3x^2 + 2x^2y + 3xy - \frac{3}{2}y^2, f(x, y) = 3x^2 + 2x^2y + 3xy - 23y^2$$

☐

$$f(x, y) = 3x^2 + 2x^2y - 3xy - \frac{3}{2}y^2, f(x, y) = 3x^2 + 2x^2y - 3xy - 23y^2$$

☐

$$f(x, y) = 3x^2 - 3xy - 2x^2y + 3y^2, f(x, y) = 3x^2 - 3xy - 2x^2y + 3y^2$$

☐

$$f(x, y) = 3x^2 + 3y^2 + 3xy, f(x, y) = 3x^2 + 3y^2 + 3xy$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$f(x, y) = 3x^2 + 2x^2y - 3xy - \frac{3}{2}y^2 \quad f(x, y) = 3x^2 + 2x^2y - 3xy - 23y^2$$

$$f(x, y) = 3x^2 - 3xy - 2x^2y + 3y^2 \quad f(x, y) = 3x^2 - 3xy - 2x^2y + 3y^2$$

$$f(x, y) = 3x^2 + 3y^2 + 3xy \quad f(x, y) = 3x^2 + 3y^2 + 3xy$$

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $f(x, y) = (P(x, y), Q(x, y))$ where $P(x, y) = 3x^2y$, $Q(x, y) = 5x + y^3$. Consider a matrix

$A = \begin{bmatrix} \frac{\partial P}{\partial x} & \frac{\partial P}{\partial y} \\ \frac{\partial Q}{\partial x} & \frac{\partial Q}{\partial y} \end{bmatrix}$. If the determinant of A is $axy^3 - bx^2$ where

$a, b \in \mathbb{R}$, then find the value of $a - b$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $f(x, y) = (P(x, y), Q(x, y))$ where $P(x, y) = 6x + y^2$, $Q(x, y) = 3x^2 + 2y$. Consider a

matrix $A = \begin{bmatrix} \frac{\partial P}{\partial x} & \frac{\partial P}{\partial y} \\ \frac{\partial Q}{\partial x} & \frac{\partial Q}{\partial y} \end{bmatrix}$. If $\det(A) = 0$, then find the value of xy

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined as $f(x, y) = e^{4(x+y)} \sin(7x - 4y)$

If $4\frac{\partial f}{\partial x} + 7\frac{\partial f}{\partial y} = k \cdot f(x, y)$ where $k \in \mathbb{R}$, then find the value of k .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 44

1 point

[Check Answers](#)

Your score is: 0/9